

# SHOPNOW

Enhancing Consumer Experience with Conversational AI &  
Virtual Try-On

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# 1. Project Overview

ShopNow is a modern e-commerce platform designed to solve the primary friction point in online fashion retail: the inability to visualize clothing fit. By integrating **IDM-VTON (Improving Diffusion Models for Authentic Virtual Try-on)** via the Hugging Face API, the system allows users to see how garments look on their own bodies in a realistic, 3D-aware environment.

## 2. Core AI Specification: Virtual Try-On

**Goal:** To provide a seamless, interactive fitting room experience using Generative AI.

### A. 4-Pose Guided Capture

Using **MediaPipe** for real-time computer vision, the system guides the user through four distinct poses (Front, Back, Left, Right). This ensures the AI model receives high-quality input with correct body alignment, minimizing distortions during the garment warping phase.

### B. IDM-VTON Pipeline (Hugging Face)

The core "try-on" logic is handled by a Diffusion-based inpainting model. The process follows these mathematical and logical steps:

- **Human Parsing:** Segmenting the user's image to isolate the torso/limbs.
- **Garment Warping:** Aligning the product image to the user's pose using a semantic feature extractor.
- **Latent Diffusion:** Synthesizing the final image where the garment follows the natural folds and lighting of the user's environment.

### 3. Technical Implementation

| Component        | Technology Stack                                      |
|------------------|-------------------------------------------------------|
| Frontend         | React, Framer Motion (Transitions), GSAP (Animations) |
| Component        | Technology Stack                                      |
| AI Engine        | Hugging Face API (IDM-VTON), MediaPipe                |
| 3D Visualization | Three.js, React Three Fiber                           |
| Backend          | Node.js, Express, Firebase (Auth/DB)                  |
| Payments         | Razorpay API Integration                              |

### 4. UI/UX & Design Philosophy

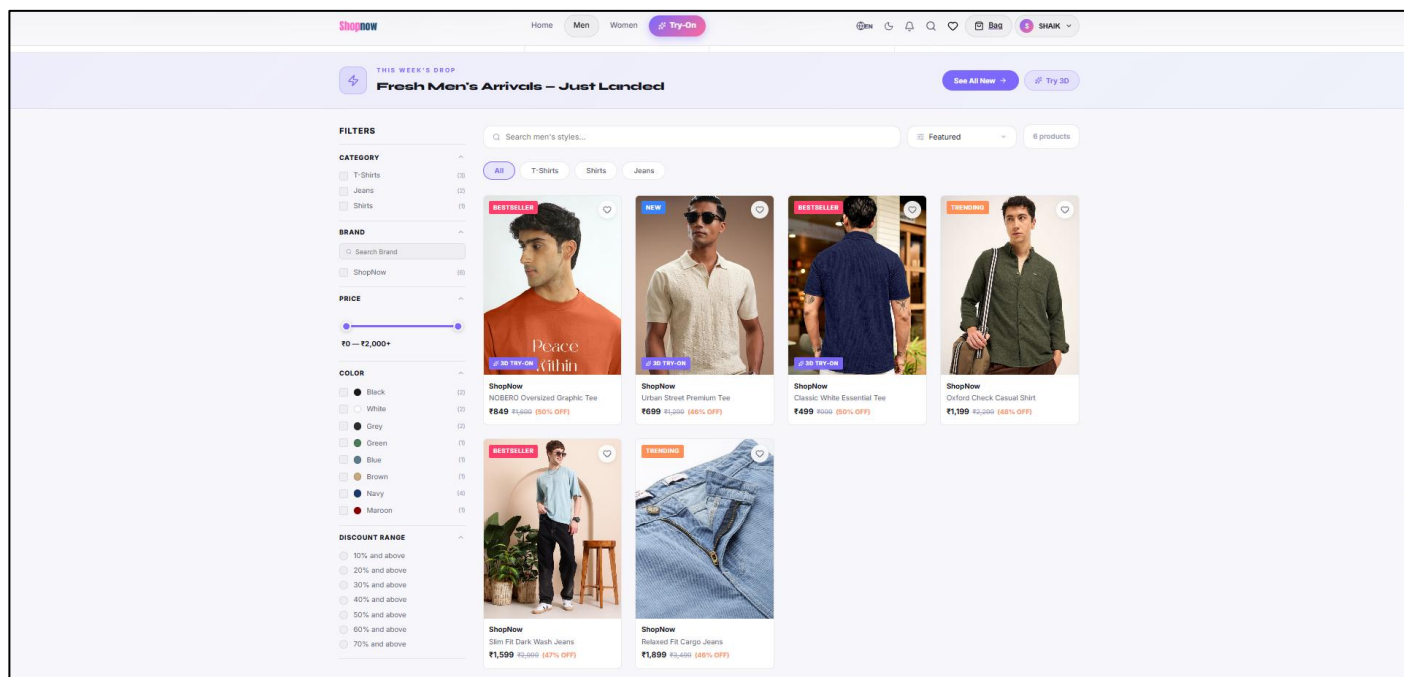
The application utilizes a **Dark Glass-morphism** design language. This aesthetic choice highlights product imagery while providing a premium, high-tech feel.

- **Smooth Navigation:** Scroll-linked parallax and GSAP animations ensure the transition between browsing and try-on is fluid.
- **Responsive Design:** Fully optimized for Mobile, Tablet, and Desktop views using Tailwind CSS media queries.

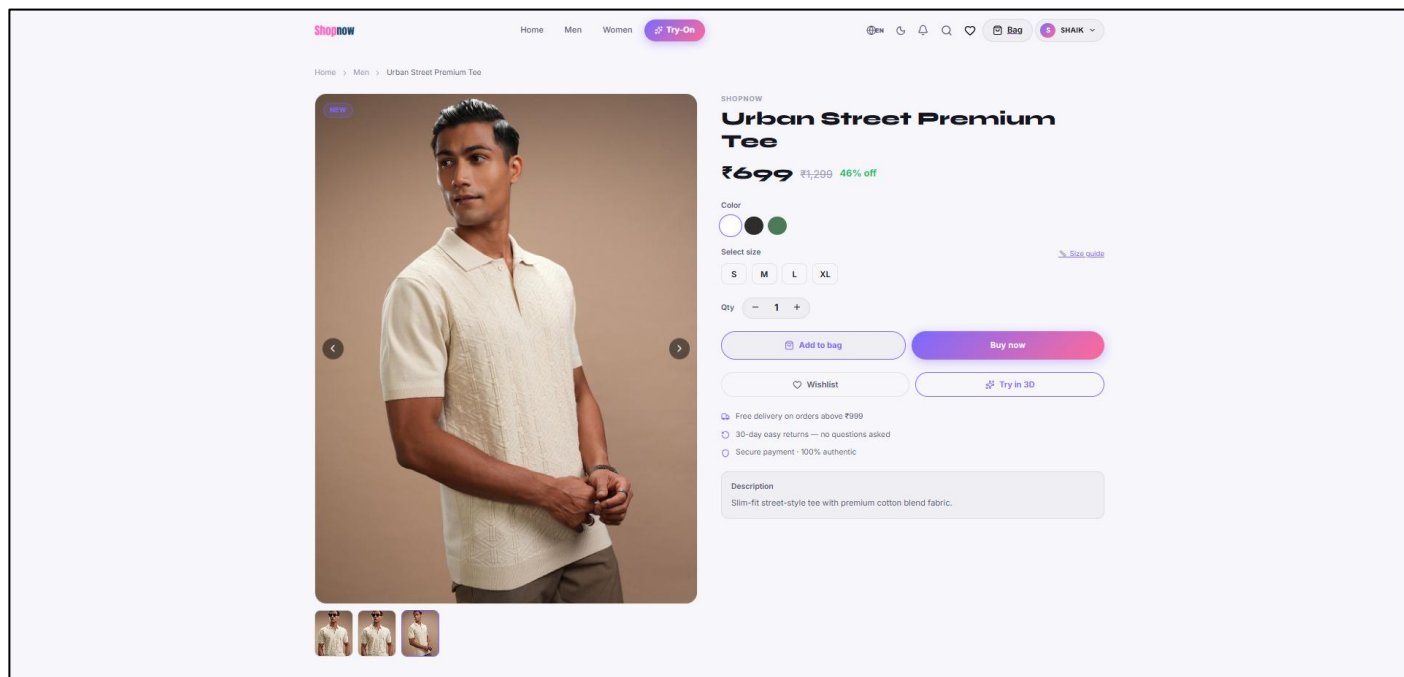
### 5. Future Roadmap

Upcoming iterations of ShopTalk will focus on **Multi-Garment Layering** (e.g., trying on a jacket over a hoodie) and **Real-time AR Mirrors**, moving beyond static image generation to full video-based virtual try-ons.

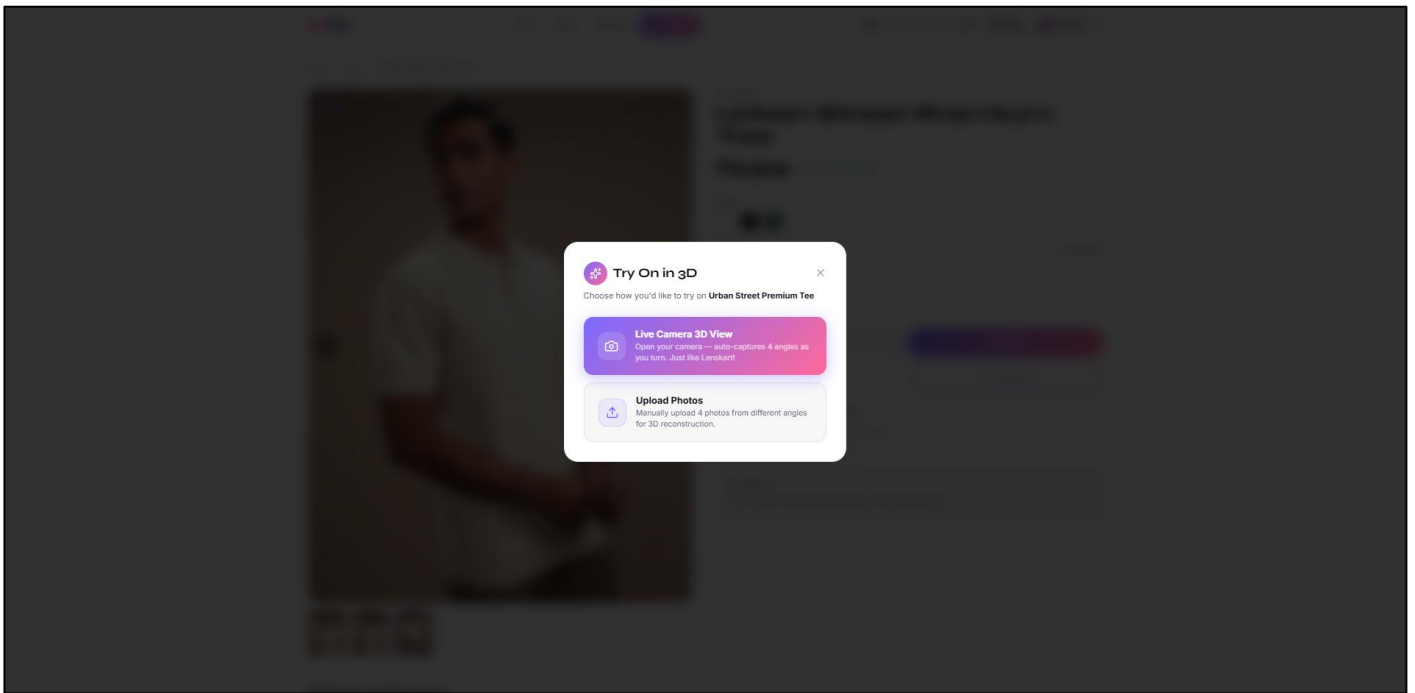
## Screenshot & Demonstration Photos:



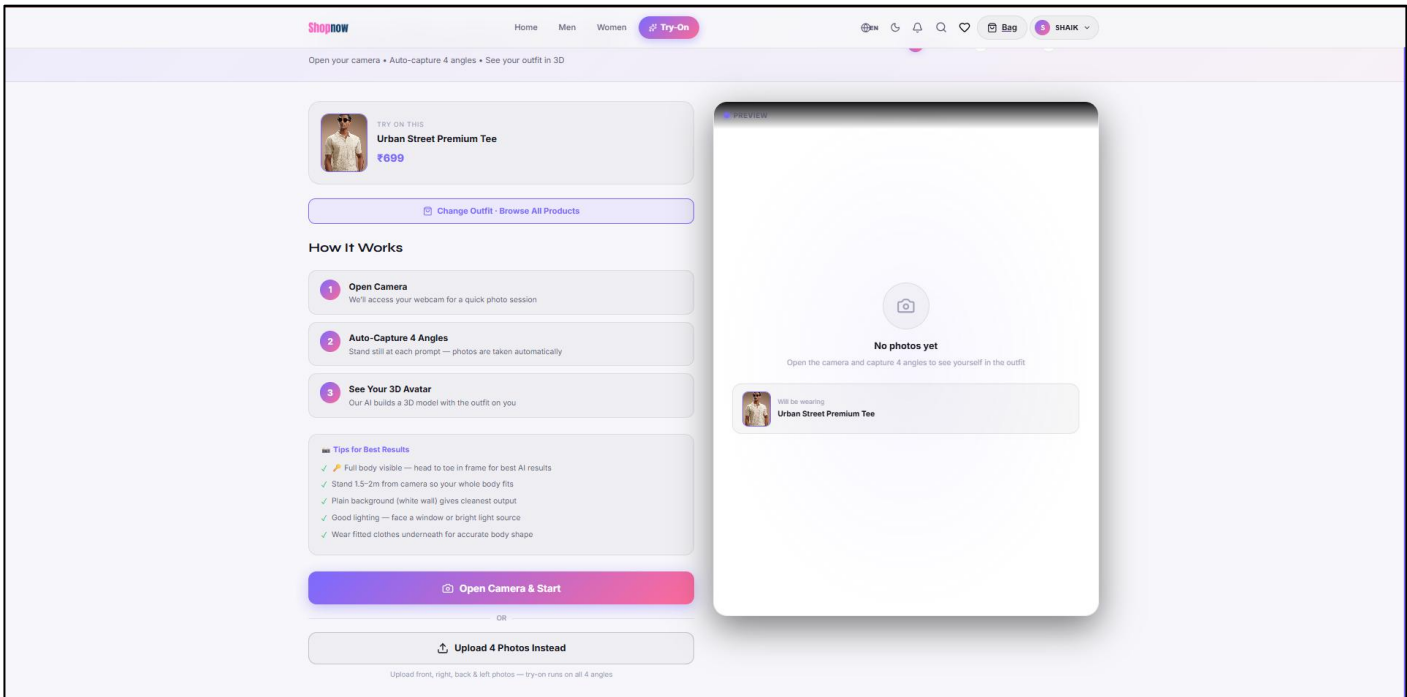
The page and product list. Users can click on any product they like to use the try-on feature



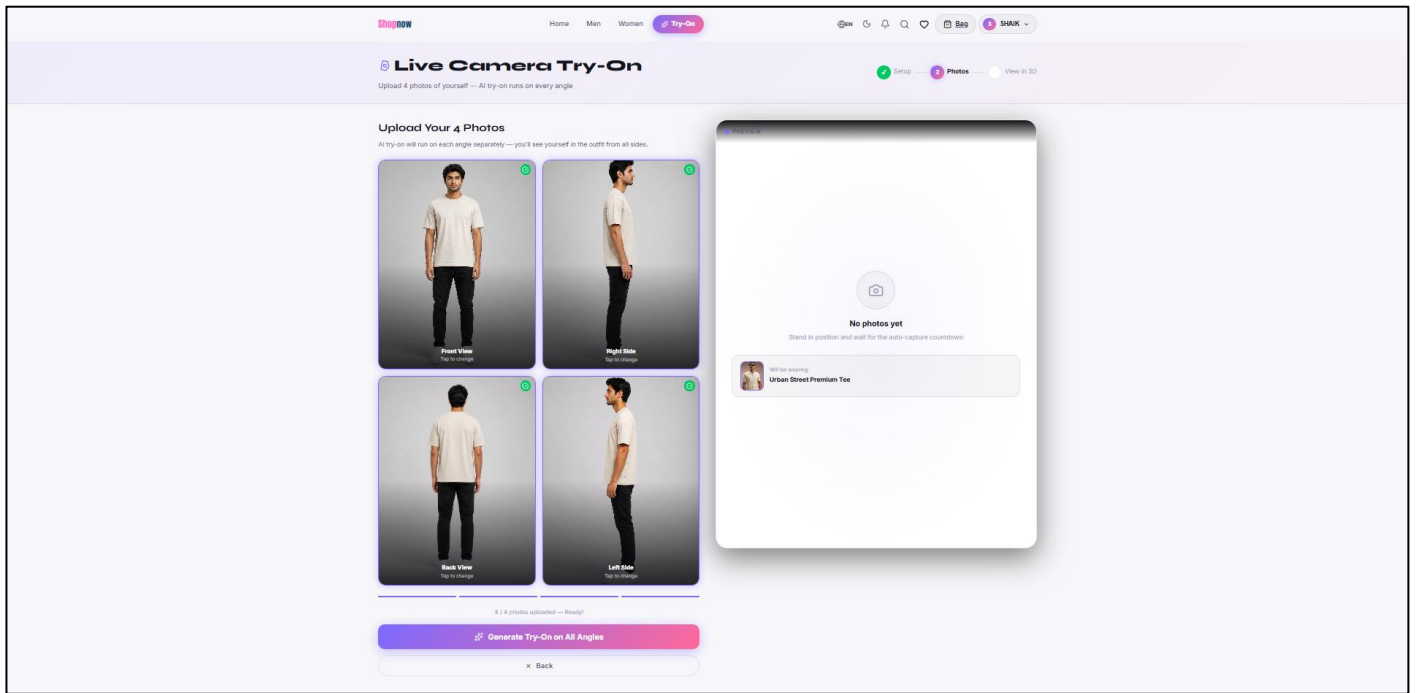
After selecting a product, you can click on the 'Try on in 3D' option. Then, by following the next steps, you can upload your photo or use your live camera to see the product in 3D.



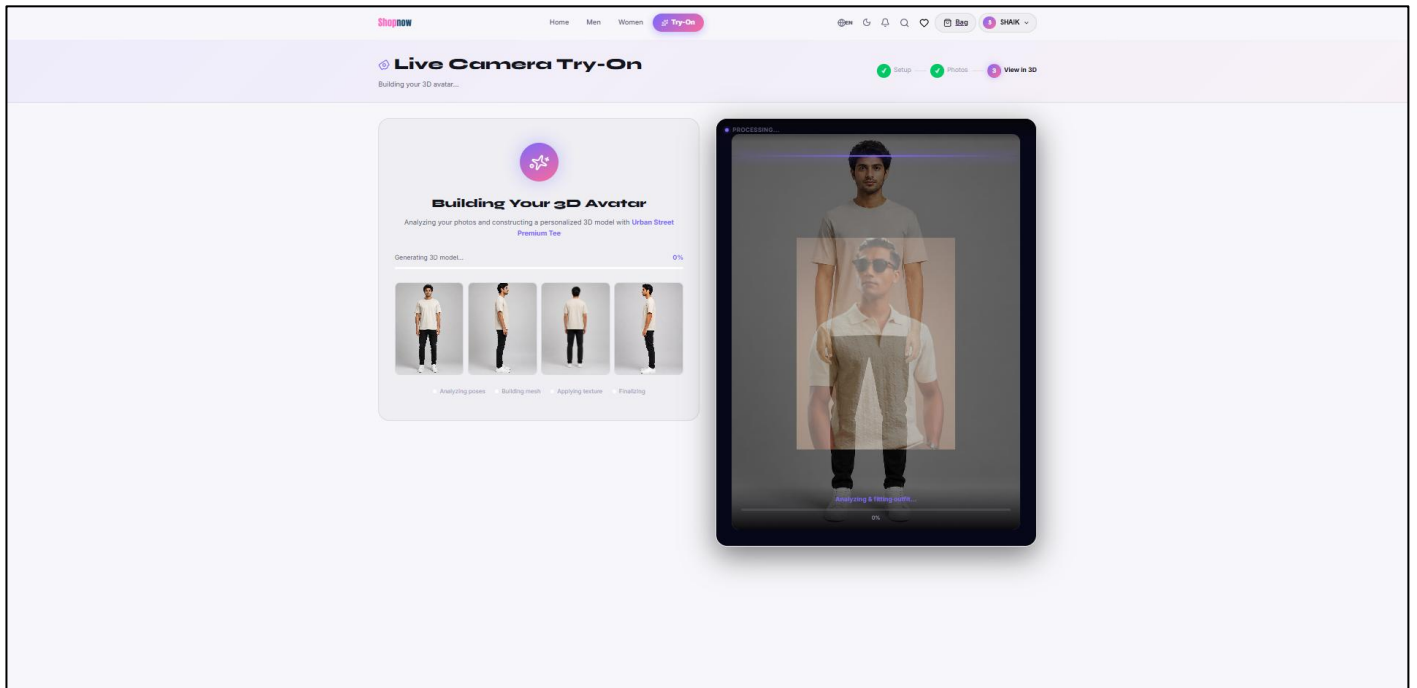
The interface now displays two options: 'Live Camera 3D View' or 'Upload Photos.' Depending on your preference, select one. If you choose the live camera, it will capture four angle photos, which you then need to upload. Alternatively, if you upload photos, you'll select four pre-existing photos taken from different angles.



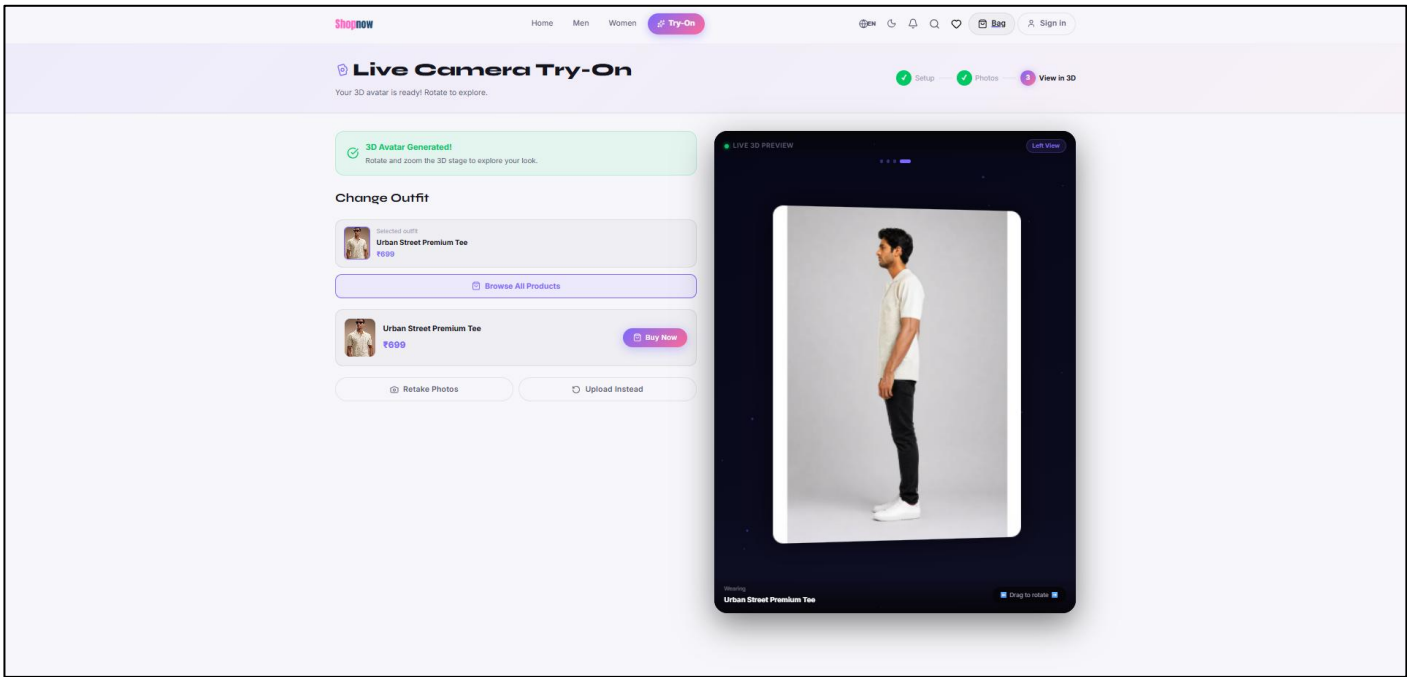
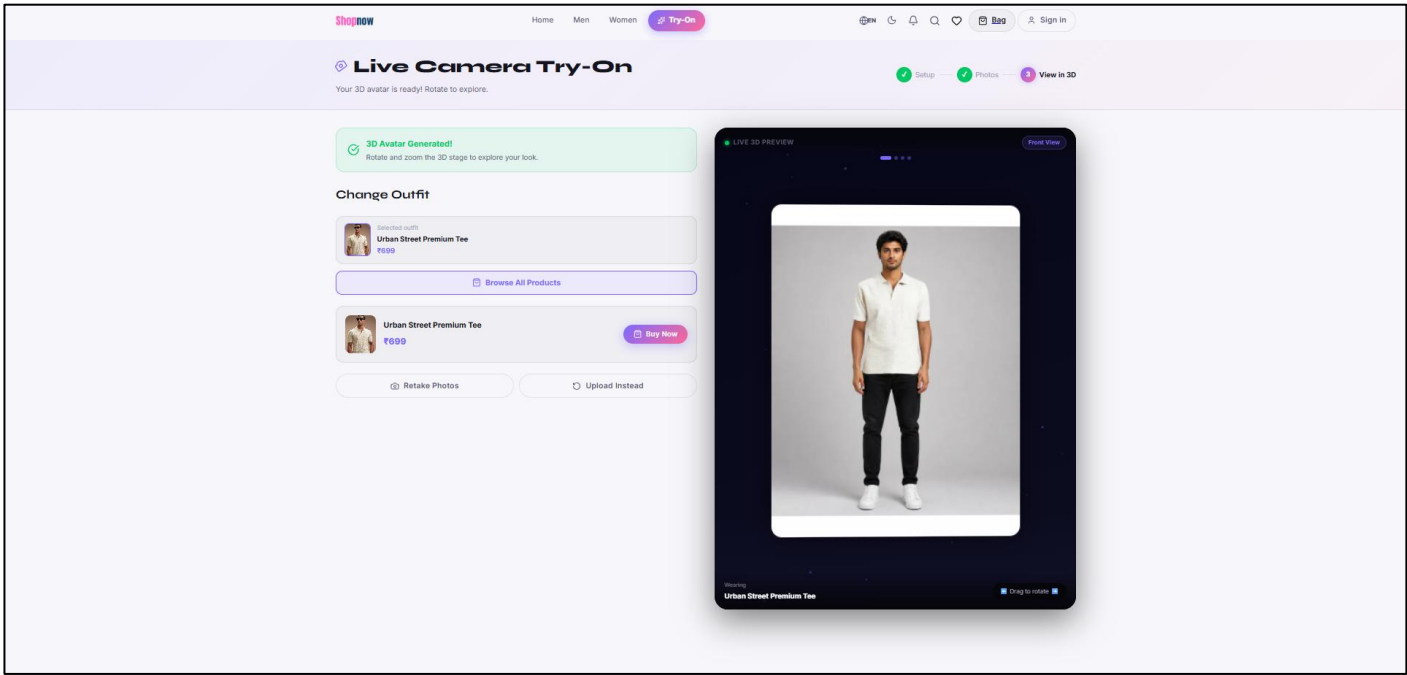
You will now see two options: 'Open Camera and Start,' where you can take new photos, and 'Upload Four Photos Instantly,' which lets you use pictures you've already saved on your device to try the feature.

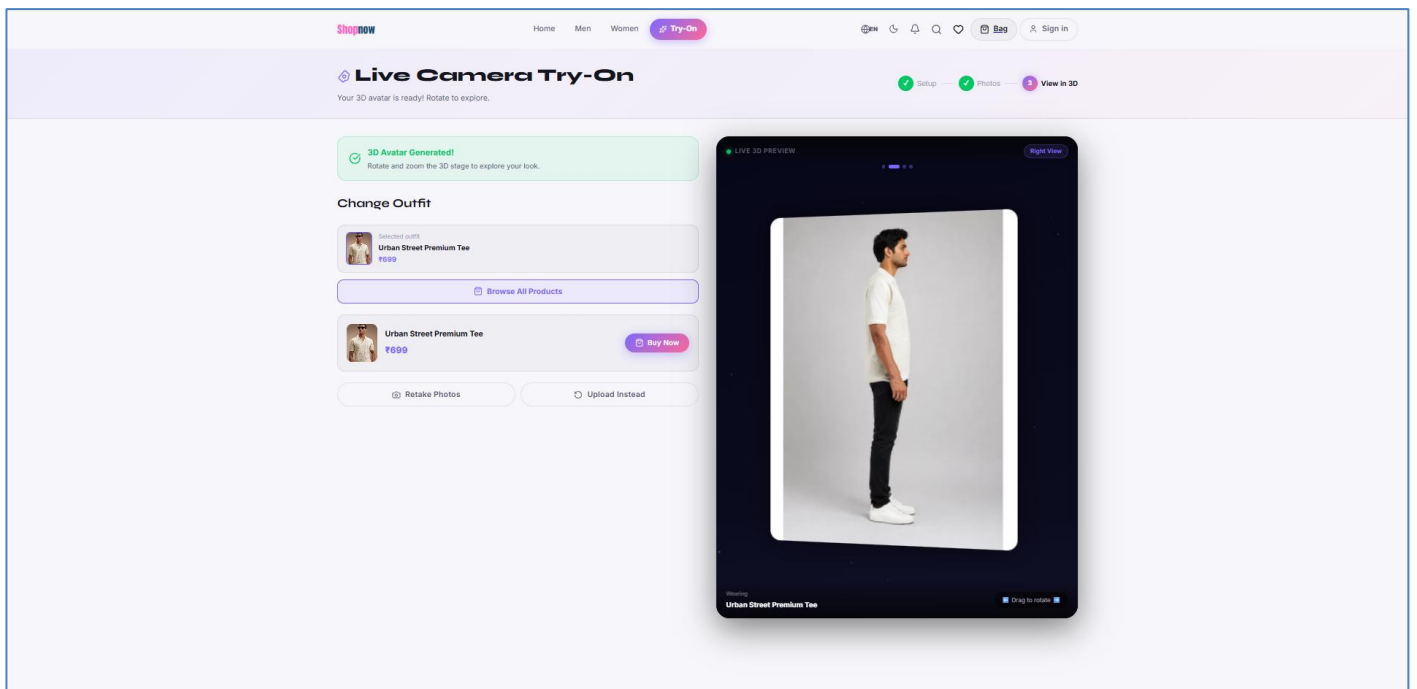
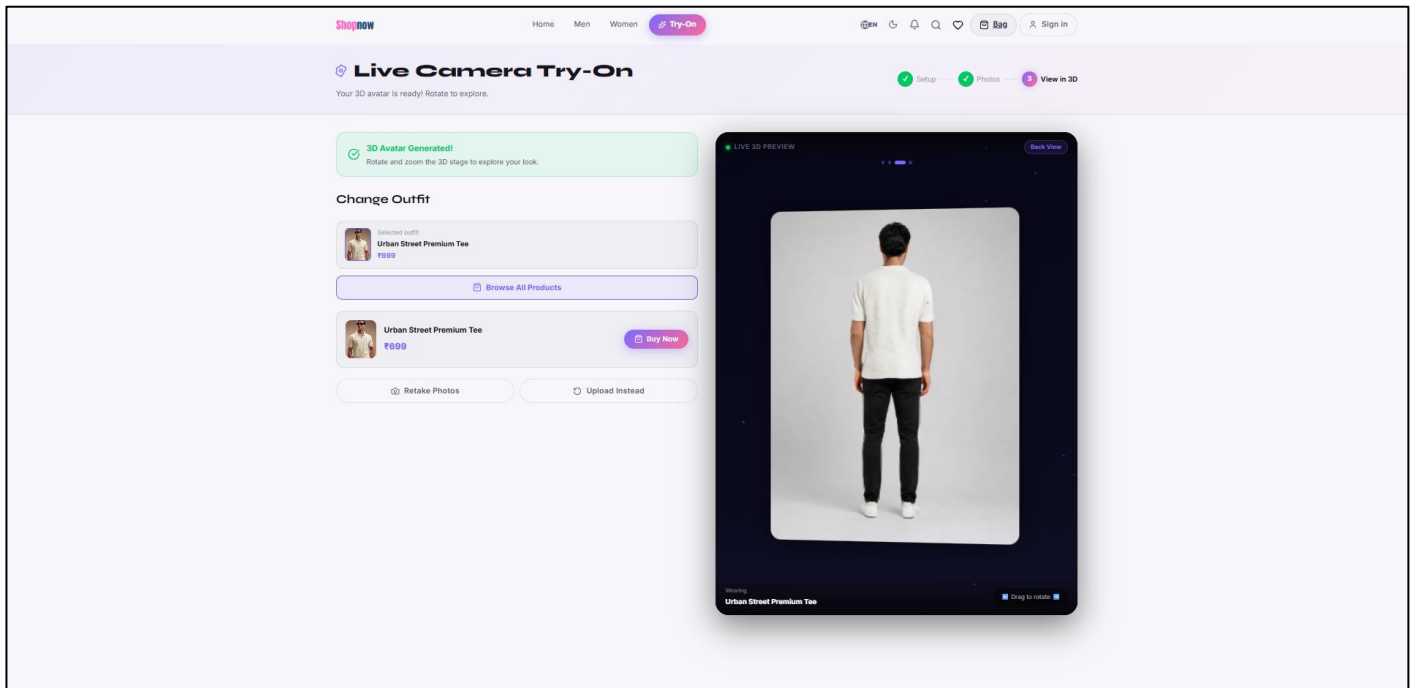


Once you've uploaded photos, either from your camera or your device, the system will analyze them and show you the results



After all photos have been uploaded, the system will verify the angles and perform a complete body analysis. This processing takes a moment to generate a proper output





Now that you've uploaded photos, the try-on feature will show you how the dress looks on you. You can try the output from front and back views, and if you like it, you can buy it